

RETAIL SALES ANALYSIS

Exploratory Data Analysis & Business Insights Report

Superstore Retail Dataset | 2015 – 2018

Data Analytics Portfolio Project

Python · Pandas · NumPy · Matplotlib · Seaborn

August 2026

Executive Summary

This report presents an end-to-end exploratory data analysis (EDA) of the Superstore retail dataset, covering 9,800 order line items across 4,922 unique orders placed between 2015 and 2018. The analysis was performed entirely in Python using Pandas for data wrangling and feature engineering, and Matplotlib / Seaborn for visualization.

The goal was to simulate a real business-intelligence workflow: clean and enrich raw transactional data, derive key performance indicators (KPIs) across products, customers, regions, and time, and translate the results into actionable insights — the kind of analysis a retail analytics or BI team would deliver to stakeholders.

\$2.26M	4,922	9,800	\$459.5
Total Sales	Total Orders	Line Items	Avg. Order Value

About the Dataset

The dataset is the widely-used “Superstore” retail sales dataset, containing order-level transaction records for a US-based office and furniture retailer. Each row represents one product line within a customer order.

Fields Used

- Order details: Order ID, Order Date, Ship Date, Ship Mode
- Customer details: Customer ID, Customer Name, Segment
- Geography: Country, City, State, Region, Postal Code
- Product details: Product ID, Product Name, Category, Sub-Category
- Metrics: Sales

Data Quality Notes

- 9,800 rows, 18 original columns; Order Date and Ship Date parsed as day-first dates.

- Postal Code had a small number of missing values (9,789 of 9,800 populated) and was not required for the analysis.
- Sales values ranged from \$0.44 to \$22,638.48 per line item, with a right-skewed distribution typical of retail transaction data.

Methodology

The analysis followed a structured workflow, moving from raw data to engineered features to grouped KPIs to visual storytelling.

1. Data Preparation

- Standardized column names (replaced spaces with underscores).
- Converted Order_Date and Ship_Date to proper datetime types (day-first format).

2. Feature Engineering

- Shipping_Days — days elapsed between order and shipment
- Order_Year, Order_Month, Month_Name, Order_Quarter, Order_Week — calendar breakdowns for trend analysis
- Day_Name and Is_Weekend — to test for weekday/weekend ordering patterns

3. KPI Computation

Over 20 grouped KPIs were computed with Pandas groupby/agg, spanning customers, products, categories, sub-categories, regions, states, segments, ship modes, and time periods — each paired with both total sales and total order counts so volume and value could be compared side by side.

4. Visualization

23 charts were produced with Matplotlib and Seaborn, including trend lines, bar charts, horizontal rankings, and heatmaps, to make each KPI immediately interpretable.

Findings: Sales Performance Over Time

Total sales grew consistently year over year, climbing from \$479,856 in 2015 to \$722,052 in 2018 — a 50.5% increase over four years — even though average order value declined slightly (from \$506.71 in 2015 to \$434.71 in 2018), meaning growth was driven primarily by a rising number of orders (947 → 1,661) rather than bigger individual purchases.

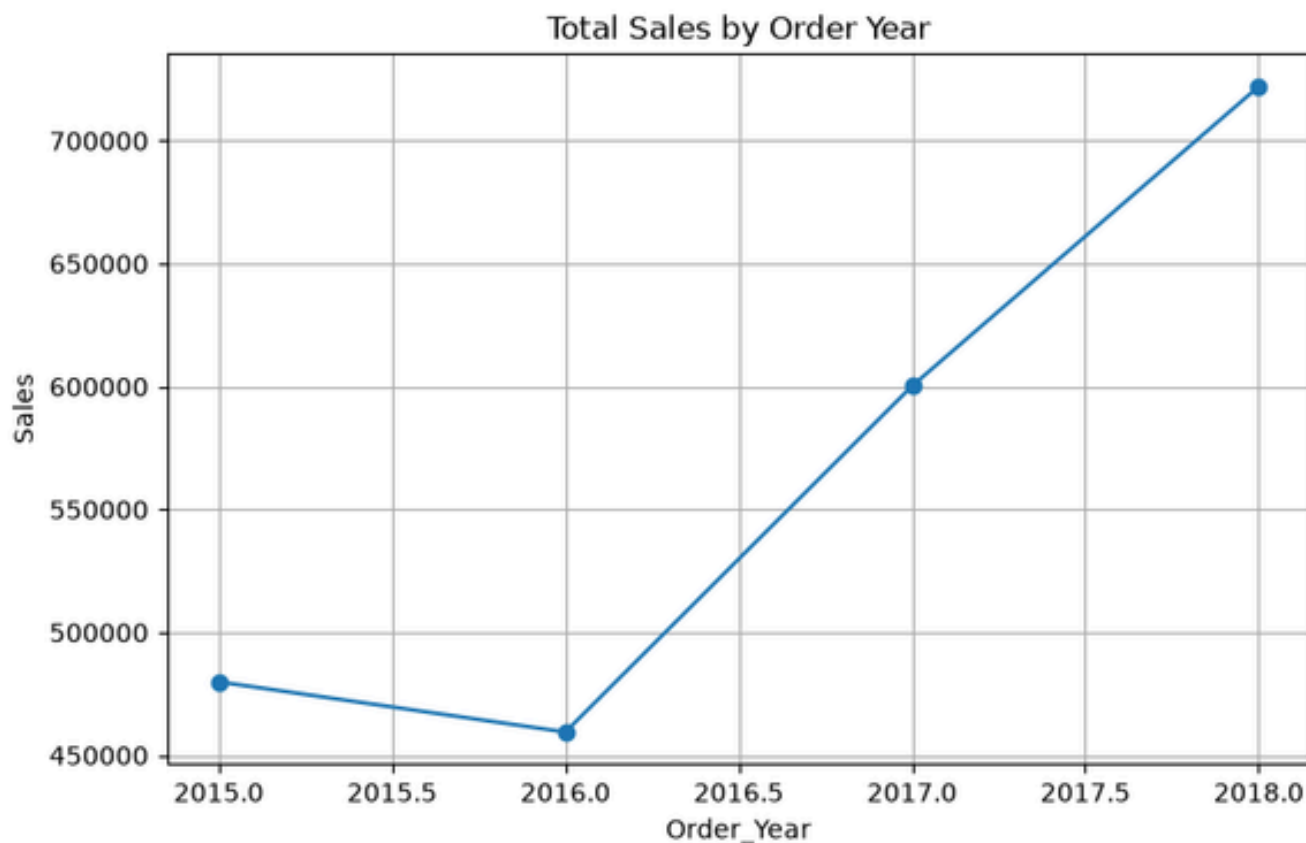


Figure 1. Total sales by order year (2015–2018)

Sales are also strongly seasonal within the year: Q4 alone accounted for \$871,138 (38.5% of total sales) versus just \$351,236 (15.5%) in Q1, and the monthly trend confirms a build-up toward November–December, consistent with holiday-season retail demand.

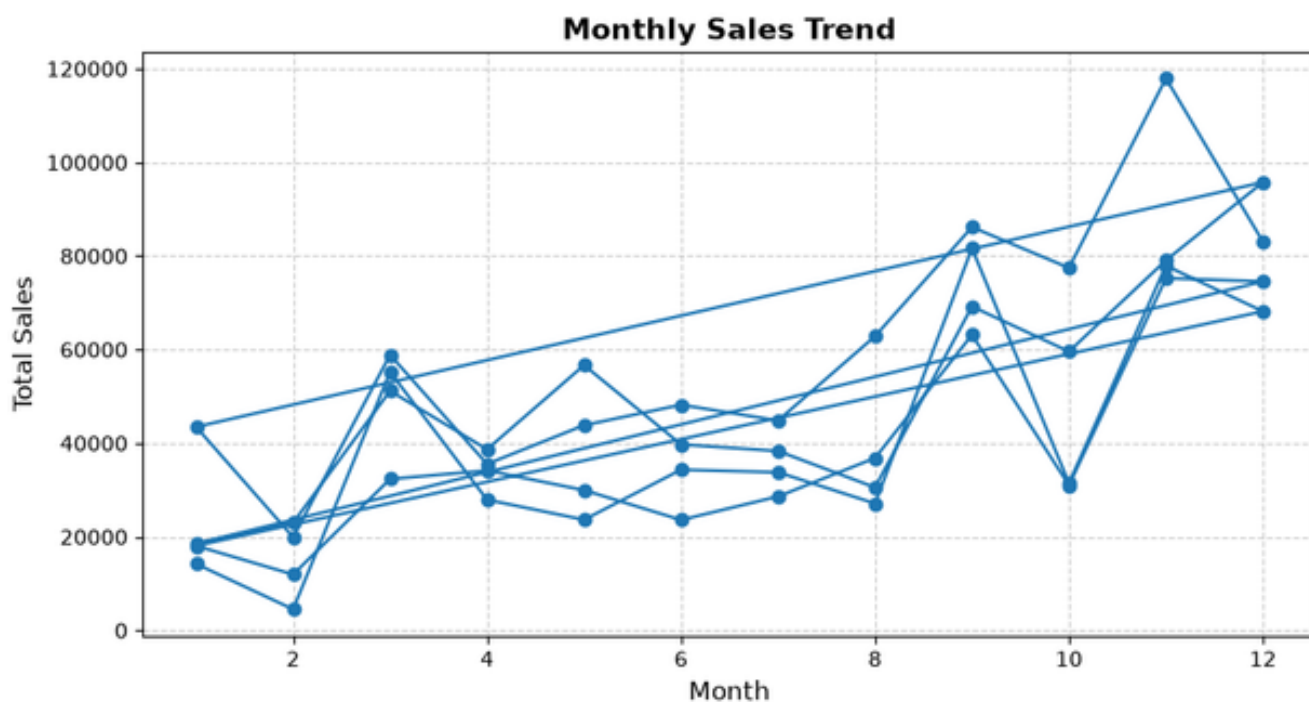


Figure 2. Monthly sales trend, aggregated across all years

Findings: Category & Product Performance

Technology generated the highest revenue (\$827,456) despite having the fewest orders (1,519), reflecting a high average transaction value driven by items like phones and machines. Office Supplies had the most orders by far (3,676) but the lowest revenue per order, typical of a high-volume, low-ticket category.

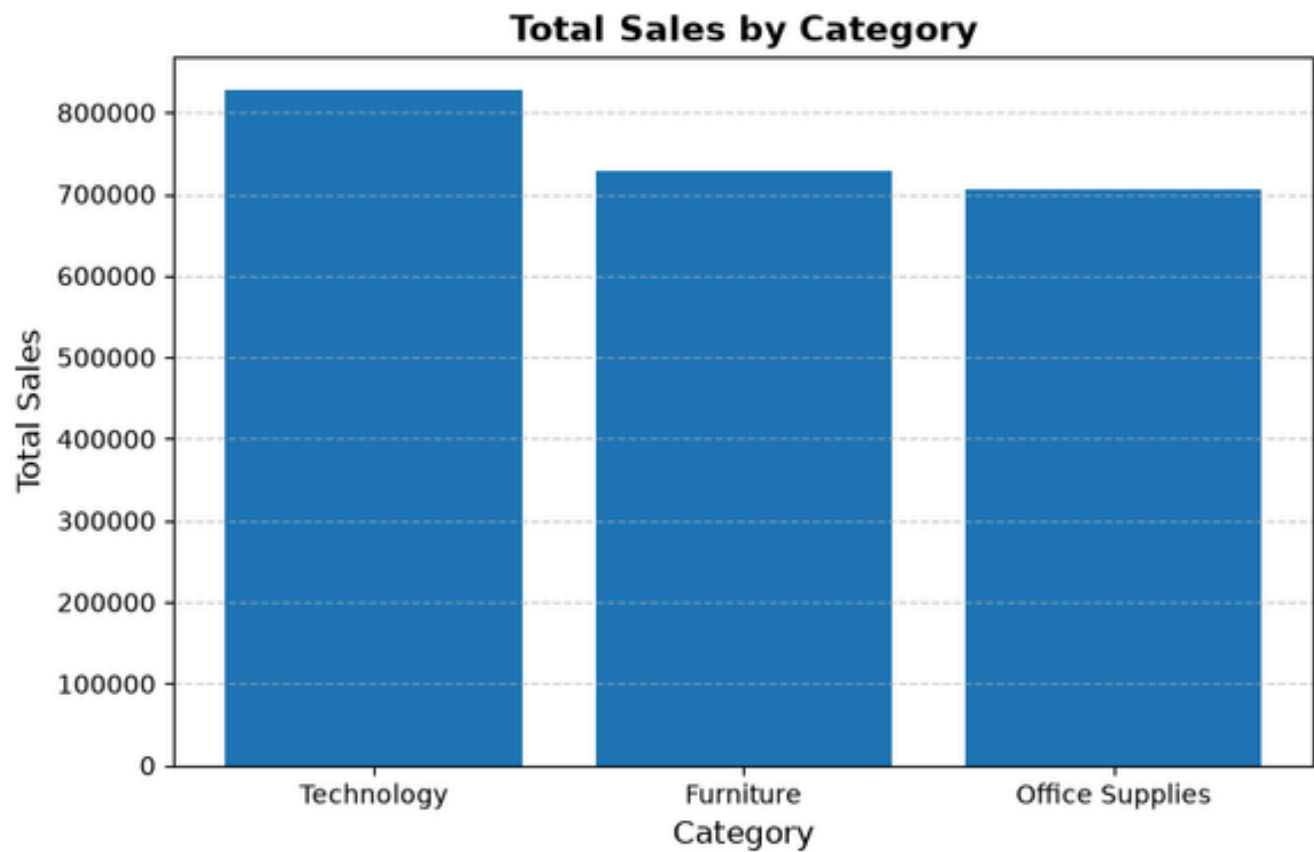


Figure 3. Total sales by product category

At the sub-category level, Phones (\$327,782) and Chairs (\$322,823) are the standout revenue drivers, followed by Storage (\$219,343) and Binders (\$200,029). Notably, Binders generated the highest order volume of any sub-category (1,291 orders) at a comparatively low average value — a classic high-frequency, low-margin product profile.

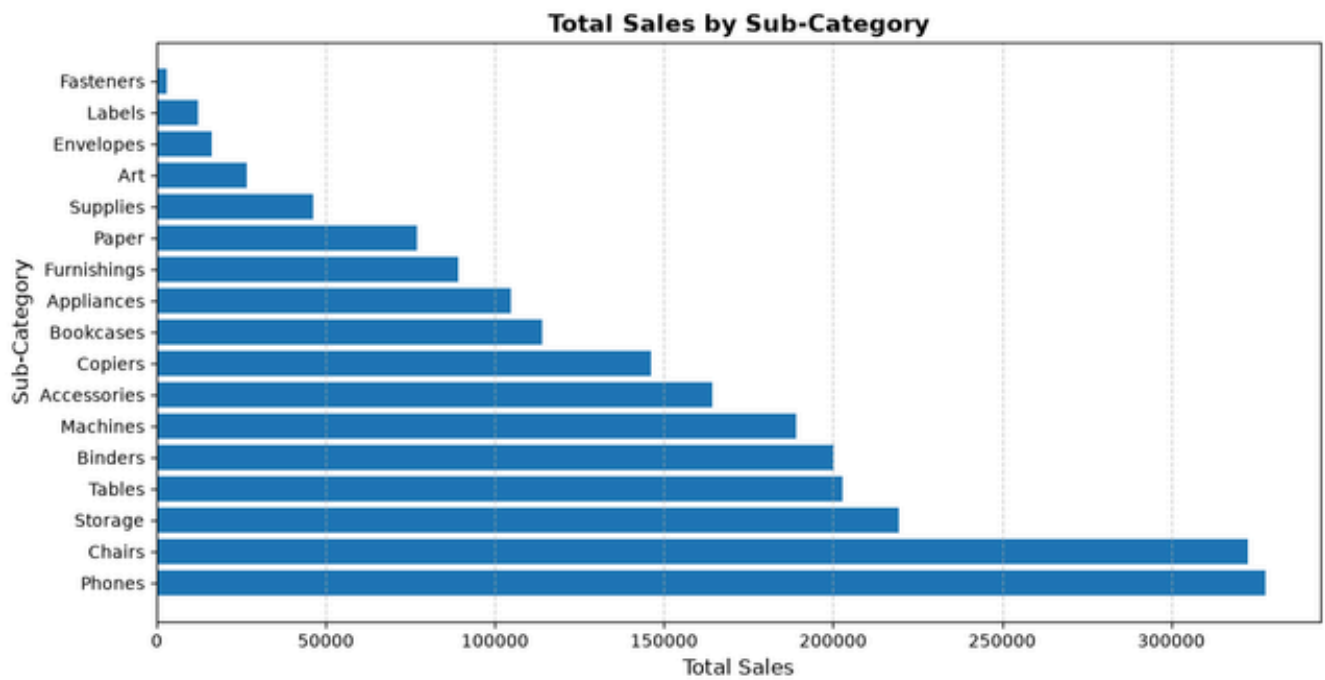


Figure 4. Total sales by sub-category, grouped by category

A small set of customers and products account for a disproportionate share of revenue. The top 10 customers alone — led by Sean Miller (\$25,043) and Tamara Chand (\$19,052) — contributed well above the average customer's lifetime spend, underlining the value of an 80/20-style customer retention strategy.

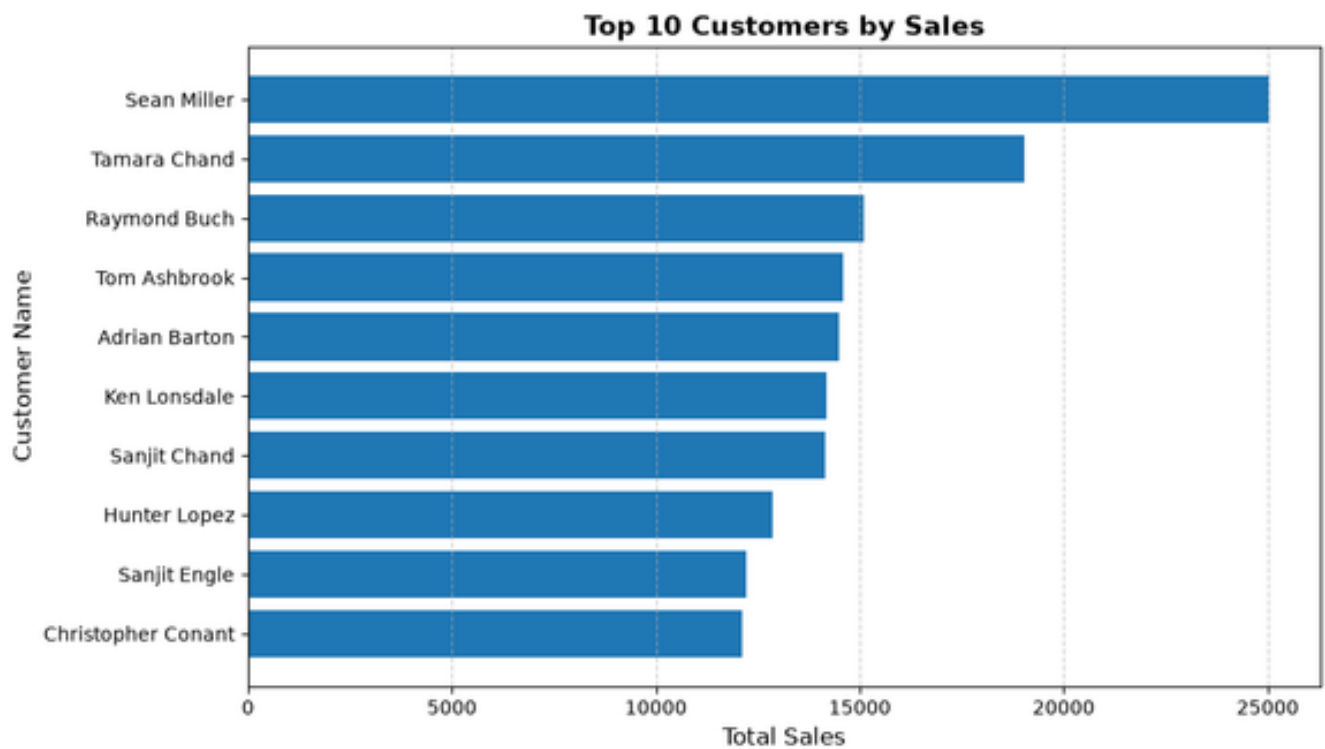


Figure 5. Top 10 customers by total sales

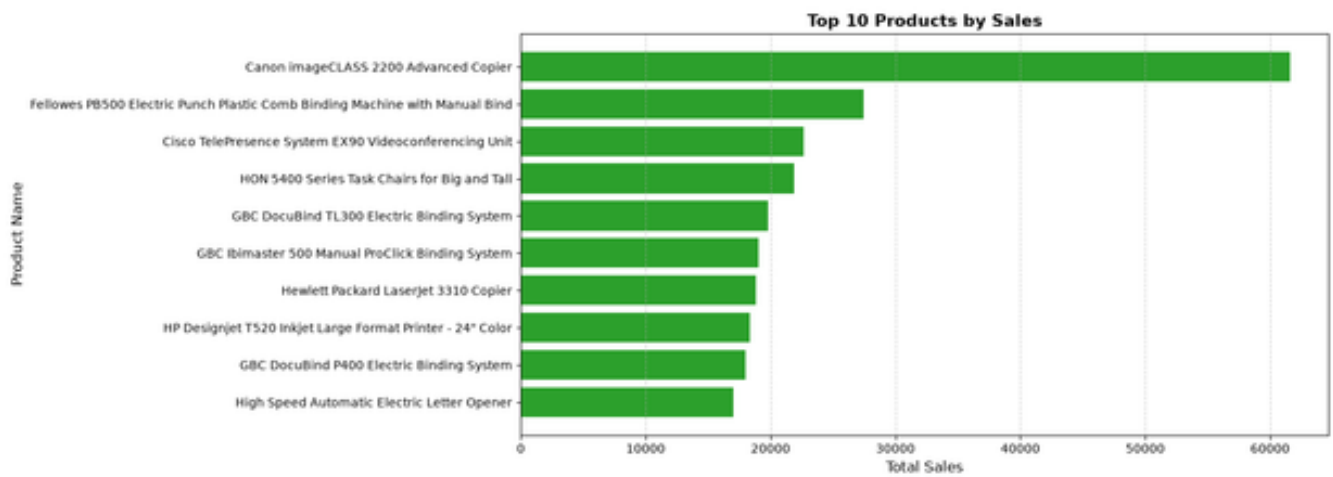


Figure 6. Top 10 products by total sales

Findings: Regional Performance

The West region led all regions in both sales (\$710,220) and order count (1,587), followed closely by the East (\$669,519 / 1,369 orders). The South region trailed on both measures (\$389,151 / 810 orders), suggesting room for expansion or a targeted regional marketing push.

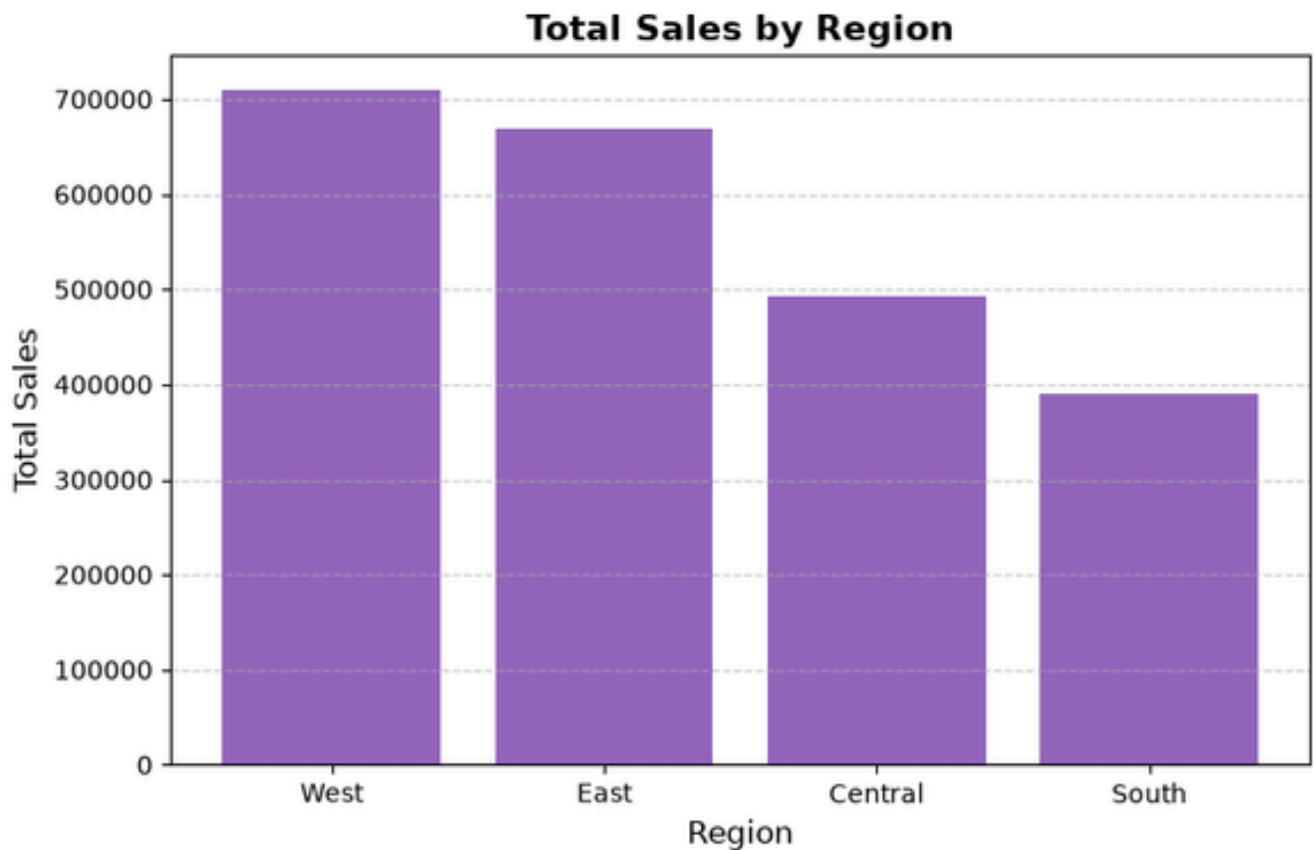


Figure 7. Total sales by region

At the state level, California (\$446,306) and New York (\$306,361) dominate, together contributing roughly a third of all national sales, while low-population states such as North Dakota, West Virginia, and Wyoming each generated under \$1,700 across the full four-year period — a pattern that closely tracks state population and commercial density.

Region vs. Category Breakdown

Cross-tabulating region against category shows that Technology and Furniture are the strongest categories in the West, while Office Supplies performs comparatively evenly across all four regions.

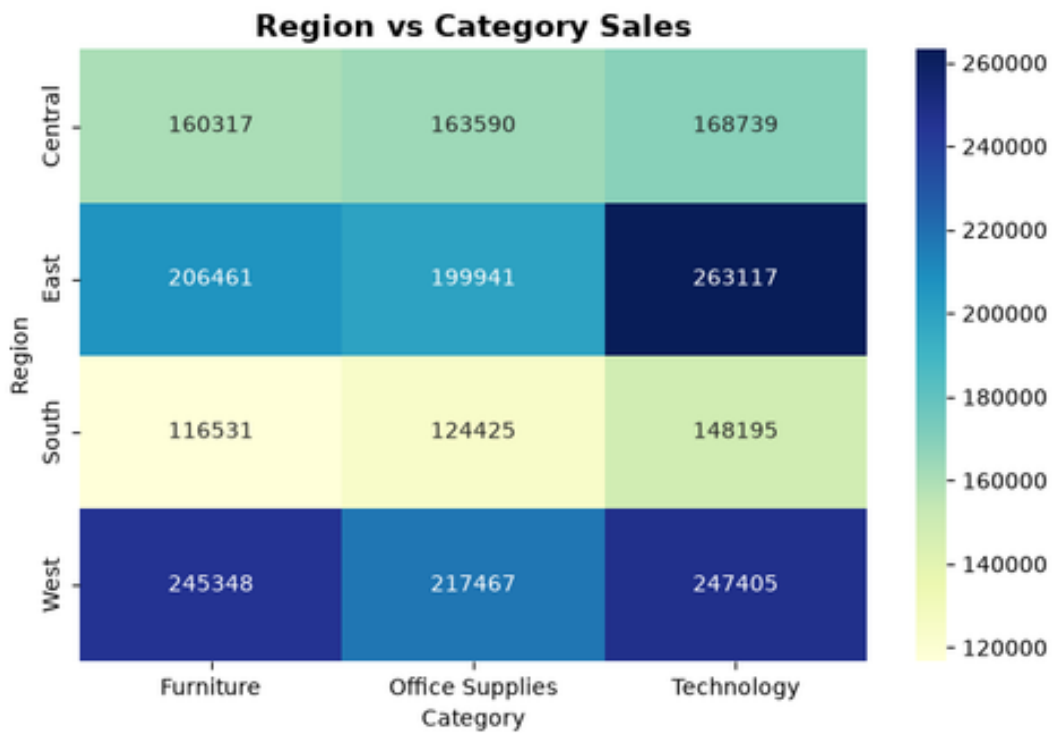


Figure 8. Sales heatmap: region vs. category

Findings: Customer Segment Analysis

The Consumer segment is the largest by both order count (2,537) and total sales (\$1.15M, 50.7% of revenue), but Home Office customers spend the most per order on average (\$475.37), followed by Corporate (\$461.77) and Consumer (\$452.53) — a gap worth exploring for upsell opportunities within the Consumer base.

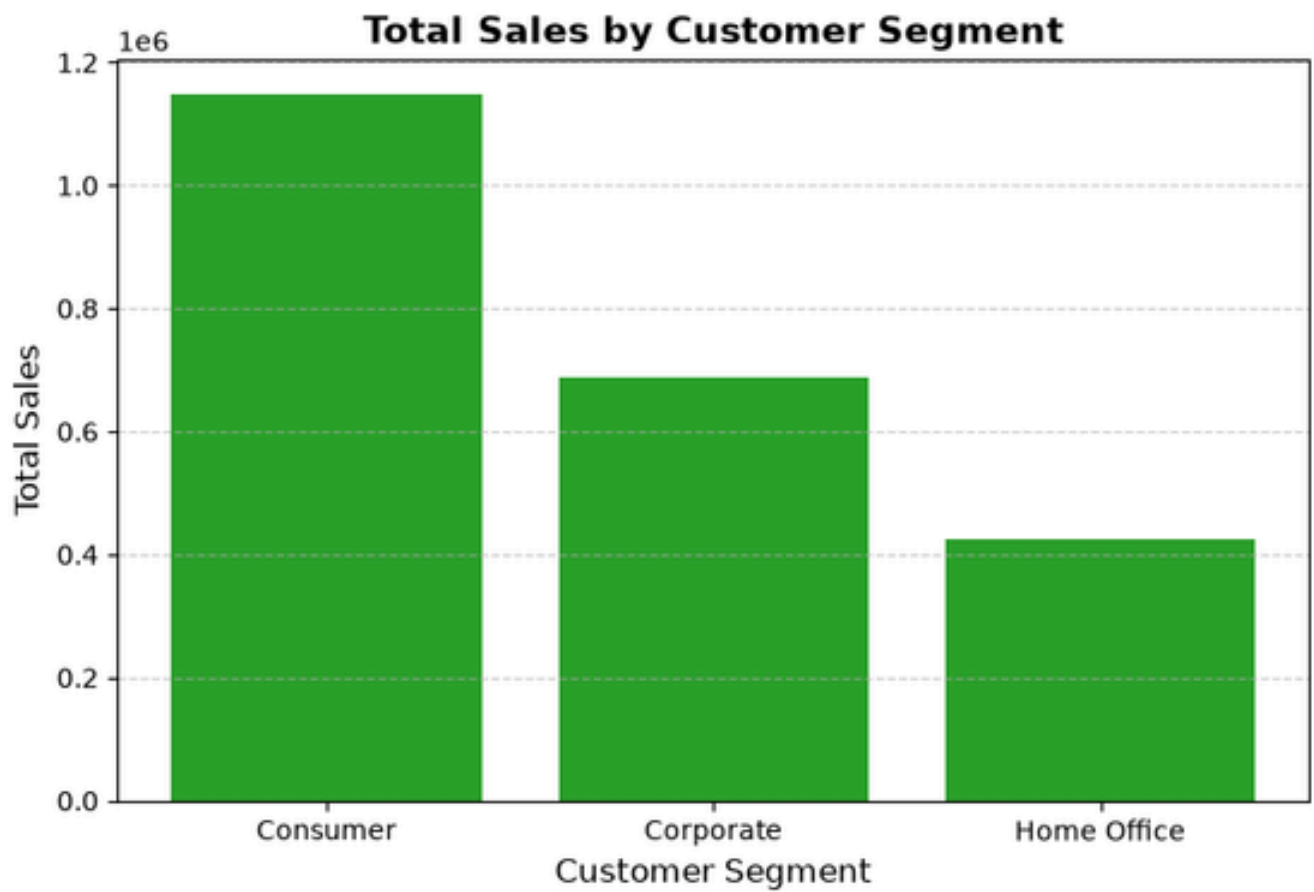


Figure 9. Total sales by customer segment

Findings: Shipping & Operations

Standard Class is by far the dominant shipping method, used in 2,945 of 4,922 orders (59.8%) and accounting for \$1.34M in sales. Average shipping time across all orders is 3.96 days, with a fairly tight spread (25th–75th percentile: 3–5 days), indicating a consistent fulfillment process regardless of order size.

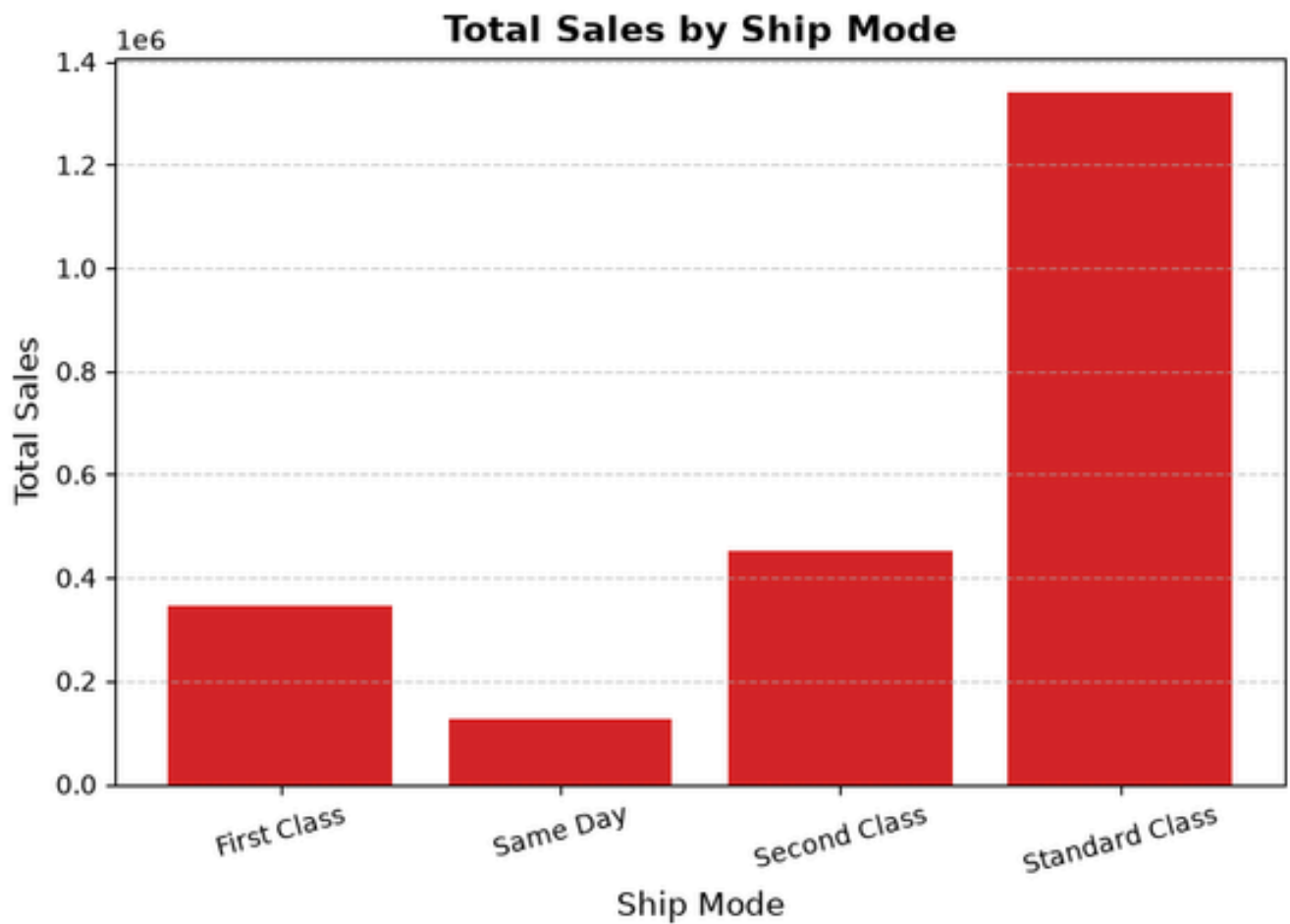


Figure 10. Total sales by ship mode

Correlation Analysis

A correlation matrix was built across Sales, Shipping_Days, Order_Year, and Order_Month to check for linear relationships that might inform forecasting. The results show negligible correlation between shipping time and sales value, and only a weak relationship between calendar timing and sales — confirming that shipping speed is not being used as a lever for higher-value orders, and that revenue growth is driven more by order volume and seasonality than by any single numeric factor in the dataset.

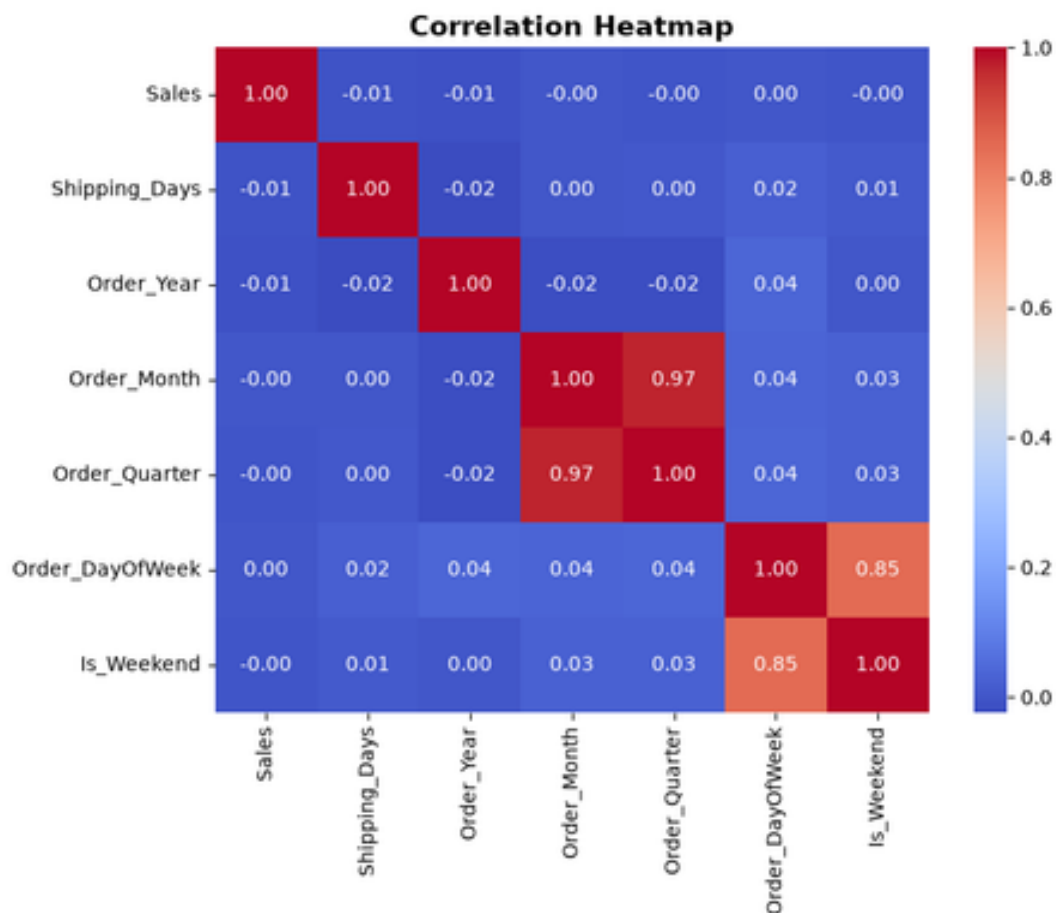


Figure 11. Correlation heatmap across numeric features

Key Insights & Recommendations

Insights

- Revenue growth (2015–2018) has been driven by order volume, not order size — average order value has trended slightly downward.
- Q4 is a critical revenue window (38.5% of annual sales), warranting inventory and staffing plans built around a holiday peak.
- Technology is the most valuable category per order; Office Supplies drives volume but at low margin potential.
- Revenue is geographically concentrated in California and New York, and functionally concentrated in a small group of top customers and products.
- Home Office customers spend more per order than Consumer or Corporate segments, despite being the smallest segment by volume.

Recommendations

- Build a Q4-specific demand-planning and marketing calendar to capitalize on the strongest sales quarter.
- Launch a retention or loyalty program targeted at top-decile customers, who contribute disproportionately to revenue.
- Investigate the South region's lower performance — whether it reflects market size, logistics, or marketing coverage — to inform expansion decisions.
- Test upsell or bundling strategies in the Consumer segment to close the average-order-value gap with Home Office customers.
- Evaluate margin, not just revenue, for high-volume/low-value sub-categories like Binders and Paper, since order volume alone can mask thin profitability.

Tools & Techniques Demonstrated

This project was built to demonstrate a practical, end-to-end data analytics skill set:

- Data cleaning and type conversion with Pandas
- Date/time feature engineering (year, month, quarter, week, weekday, weekend flags)
- Grouped aggregation and multi-key KPI design (groupby, agg, merge)
- Ranking and outlier analysis (nlargest / nsmallest for top and bottom performers)
- Data visualization with Matplotlib (line, bar, horizontal bar charts) and Seaborn (heatmaps)
- Correlation analysis for identifying (or ruling out) linear relationships between metrics
- Structuring analysis around business KPIs rather than purely technical exploration

Conclusion

This analysis turned a raw 9,800-row transactional export into a structured set of business KPIs and visual insights covering sales trends, product and customer performance, regional dynamics, and operational metrics. The workflow mirrors how a retail analytics team would approach a quarterly or annual business review, and the resulting recommendations — from Q4 planning to customer retention to regional investment — are the kind of concrete, data-backed next steps a stakeholder could act on directly.